

# Scout Engine Scoring Matrix

Multi-timeframe RSI cumulative breakout detection framework designed for early recovery identification in Indian equities. Part one of a two-engine system: Scout for early detection, Sentinel for confirmed trend entry.

Pool A: 45 pts | Pool B: 25 pts | Pool C: 15 pts  
Bonus: 25 pts | Core Cap: 85 | Max Score: 110

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>Architecture Overview</b>                       | <b>2</b>  |
| Two-Engine Architecture . . . . .                  | 2         |
| Key Design Changes from v2 . . . . .               | 2         |
| Timeframes . . . . .                               | 3         |
| Formula and Scoring Rules . . . . .                | 3         |
| <b>Pool A - RSI Recovery   45 pts</b>              | <b>4</b>  |
| Signal Table . . . . .                             | 4         |
| RSI Reversal Formula . . . . .                     | 5         |
| RSI Recovery Momentum Formula . . . . .            | 5         |
| Multi-Speed RSI Divergence . . . . .               | 6         |
| <b>Pool B - Price Structure   25 pts</b>           | <b>6</b>  |
| Signal Table . . . . .                             | 6         |
| Price Approaching TP - Proximity Scaling . . . . . | 7         |
| <b>Pool C - Confirmation   15 pts</b>              | <b>7</b>  |
| <b>Bonus Pool   25 pts</b>                         | <b>8</b>  |
| Bonus A - Structural   12 pts . . . . .            | 8         |
| Bonus B - Uptrend Correction   13 pts . . . . .    | 8         |
| <b>Entry Decision Thresholds</b>                   | <b>9</b>  |
| <b>Trough-Gated Architecture</b>                   | <b>10</b> |
| <b>Signal Lifecycle Definitions</b>                | <b>11</b> |
| <b>Design Principles</b>                           | <b>12</b> |
| No Inter-Pool Gating . . . . .                     | 12        |
| Per-Timeframe Independence . . . . .               | 12        |
| Timeframe Weighting . . . . .                      | 12        |
| Early-Firing Signal Philosophy . . . . .           | 12        |
| Sectoral Relativity . . . . .                      | 13        |
| <b>Consolidated Weight Summary</b>                 | <b>13</b> |

# Architecture Overview

The Scout Engine is the first of a two-engine architecture designed for multi-timeframe RSI cumulative breakout detection in Indian equities. It is purpose-built for **early detection** of recovery signals that fire before price confirms a breakout. The companion **Sentinel Engine** (to be designed separately) handles confirmed-trend entries with heavier price-structure and confirmation weighting. Together, these two engines cover the full spectrum from early reconnaissance to validated trend-continuation entries, ensuring that no actionable signal is missed regardless of where a stock sits in its recovery cycle.

The core problem the Scout Engine solves is straightforward: the v2 scoring matrix allocated approximately 55 of its 115 core points behind late-firing signals such as Price > TP28, All RSI > 50, and TP > WC50. These signals only activate after price has already crossed the breakout level, defeating the purpose of early entry. For Apollo Microsystems, this meant the v2 matrix could not reach a score of 75 until price had already surpassed 235. The Scout Engine redistributes weight toward RSI recovery dynamics (Pool A) and introduces new early-firing signals that score points while price is still basing below the breakout zone.

## Two-Engine Architecture

The system uses two distinct scoring engines, each optimized for a different phase of the recovery cycle. The Scout Engine activates first, detecting early RSI recovery patterns and structural basing signals to identify high-probability entry points before the crowd. Once the Scout Engine has flagged a stock and price begins to confirm with structure, the Sentinel Engine takes over with heavier confirmation weighting for trend-continuation entries. A stock can be scored by both engines simultaneously; they serve different use cases rather than representing a sequential upgrade path.

| Attribute                | Scout Engine                   | Sentinel Engine                 |
|--------------------------|--------------------------------|---------------------------------|
| Purpose                  | Early detection below breakout | Confirmed trend continuation    |
| Pool A (RSI Recovery)    | 45 pts (Heavy)                 | 25-30 pts (Standard)            |
| Pool B (Price Structure) | 25 pts (Moderate)              | 30-35 pts (Heavy)               |
| Pool C (Confirmation)    | 15 pts (Light)                 | 25-30 pts (Heavy)               |
| Core Cap                 | 85                             | 90-95                           |
| Bonus Pool               | 25 pts                         | 25-30 pts                       |
| Max Possible Score       | 110                            | 120-125                         |
| Target Entry Zone        | Below breakout price           | Above breakout, trend confirmed |
| Signal Character         | Recovery momentum, divergence  | Structure, volume, confirmation |

## Key Design Changes from v2

| Aspect                             | v2 Matrix                     | Scout Engine                                         |
|------------------------------------|-------------------------------|------------------------------------------------------|
| <b>Timeframes</b>                  | 4H, D, 3D, W (4 TFs)          | 4H, D, W (3 TFs; no 3D data)                         |
| <b>Pool A Weight</b>               | 55 pts                        | 45 pts (streamlined)                                 |
| <b>Pool B Weight</b>               | 35 pts                        | 25 pts (reduced late-firing signals)                 |
| <b>Pool C Weight</b>               | 25 pts                        | 15 pts (lighter confirmation)                        |
| <b>Core Cap</b>                    | 100                           | 85 (sum of pool maxima)                              |
| <b>Price &gt; TP Signal</b>        | 13 pts, requires Close > TP28 | Replaced by "Price Approaching TP" (proximity-based) |
| <b>All RSI &gt; 50</b>             | 6.5 pts (late-firing)         | Removed from Scout; captured via Recovery Momentum   |
| <b>RSI Recovery Momentum</b>       | Not present                   | NEW: 7 pts max, rate-based early detection           |
| <b>Cross-TF Recovery Alignment</b> | Not present                   | NEW: 4 pts, multi-TF recovery agreement              |
| <b>Multi-TF Trough Agreement</b>   | Not present                   | NEW: 2.5 pts, troughs across timeframes              |
| <b>D-RSI21 &gt; 60</b>             | 5 pts (late-firing)           | Replaced by D-RSI21 > 50 (earlier activation)        |
| <b>Inter-Pool Gating</b>           | None                          | None (retained)                                      |

## Timeframes

The Scout Engine operates on **three timeframes**: 4-Hour (4H), Daily (D), and Weekly (W). The 3-Day timeframe from the v2 matrix has been removed because the user data pipeline provides separate CSV files only for 4H, Daily, and Weekly intervals. Each timeframe is scored independently; lower timeframes lead higher timeframes in the recovery sequence. A bullish crossover on the 4H RSI typically fires days before the equivalent signal appears on the Daily chart, and the Weekly RSI is the last to confirm. This natural cascading behavior is what allows the Scout Engine to accumulate points early in the recovery cycle.

## Formula and Scoring Rules

$$\text{Entry Score} = \min(85, \text{Pool A} + \text{Pool B} + \text{Pool C}) + \text{Bonus A} + \text{Bonus B}$$

Core pools (A + B + C) are capped at 85, which is the sum of their individual pool maxima (45 + 25 + 15). In practice, the cap rarely activates because it would require every single signal across all pools and timeframes to fire simultaneously. Bonus points (A + B) are additive above the core cap, rewarding structural advantages such as volume surges, proximity to support, and secular uptrend confirmation. The maximum possible score is 110 (85 core + 25 bonus). Position sizing scales with total score as

defined in the threshold table.

## Pool A - RSI Recovery | 45 pts

Pool A is the backbone of the Scout Engine, carrying the heaviest weight at 45 points. It tracks individual RSI recovery dynamics, bullish crossovers, and divergence patterns across all three timeframes. Points grow as RSIs recover from their cycle lows and shrink if RSIs fall back, reflecting the dynamic nature of momentum recovery. Each signal on each timeframe is scored independently, meaning a 4H RSI21 crossover contributes its points regardless of whether the Daily or Weekly crossover has occurred.

The key innovation in the Scout Engine Pool A is the introduction of **RSI Recovery Momentum**, a new signal that measures the rate of RSI rise rather than absolute RSI level. This fires early in the recovery cycle when RSI is accelerating upward from oversold conditions, even if it has not yet crossed the 50 level. Similarly, **Cross-TF Recovery Alignment** and **Multi-TF Trough Agreement** are new aggregate signals that reward multi-timeframe confluence without requiring any single timeframe to have completed its full recovery.

### Signal Table

| Signal                            | Condition                                                                                                   | 4H    | D     | W     | Total | Lifecycle     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|---------------|
| <b>Multi-Speed RSI Divergence</b> | Price LL, RSI HL at local minima. Weight scales with number of RSI periods (21, 36, 56) showing divergence. | 0.5   | 1.0   | 2.0   | 3.5   | Fire-and-hold |
| <b>RSI21 Reversal</b>             | Recovery from cycle low, scaled 0-100%. Power-curve scaling: faster reward early.                           | 0-1.5 | 0-2.5 | 0-3.5 | 0-7.5 | Dynamic       |
| <b>RSI36 Reversal</b>             | Recovery from cycle low, scaled 0-100%. Power-curve scaling.                                                | 0-0.5 | 0-1.0 | 0-1.5 | 0-3.0 | Dynamic       |
| <b>RSI56 Reversal</b>             | Recovery from cycle low, scaled 0-100%. Power-curve scaling.                                                | 0-0.5 | 0-0.5 | 0-1.0 | 0-2.0 | Dynamic       |
| <b>RSI21 &gt; RSI36</b>           | Bullish crossover: RSI21 crosses above RSI36.                                                               | 1.5   | 2.5   | 3.5   | 7.5   | Live/die      |
| <b>RSI21 &gt; RSI56</b>           | Bullish crossover: RSI21 crosses above RSI56.                                                               | 1.0   | 1.5   | 2.5   | 5.0   | Live/die      |
| <b>RSI36 &gt; RSI56</b>           | Bullish crossover: RSI36 crosses above RSI56.                                                               | 0.5   | 1.0   | 1.5   | 3.0   | Live/die      |

| Signal                             | Condition                                                                                                                             | 4H    | D     | W     | Total | Lifecycle     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------|-------|-------|-------|---------------|
| <b>RSI Recovery Momentum</b>       | NEW. Rate of RSI21 recovery: (current - 3 bars ago) / 3. Scaled by slope magnitude. Fires when RSI is actively rising, even below 50. | 0-1.5 | 0-2.5 | 0-3.0 | 0-7.0 | Dynamic       |
| <b>Cross-TF Recovery Alignment</b> | NEW. 2+ TFs showing RSI21 positive momentum simultaneously. 2 TFs = 2 pts, 3 TFs = 4 pts. Single scalar.                              | -     | -     | -     | 4.0   | Live/die      |
| <b>Multi-TF Trough Agreement</b>   | NEW. Troughs detected on 2+ TFs within a 5-bar window. 2 TFs = 1.5 pts, 3 TFs = 2.5 pts. Single scalar.                               | -     | -     | -     | 2.5   | Fire-and-hold |

Raw maximum across all signals: 45.0 points. Each signal fires independently on each timeframe. Dynamic signals (Reversal, Recovery Momentum) scale continuously from 0 to their maximum as RSI moves. Live/die signals activate on condition met and deactivate if the condition fails sustainably. Fire-and-hold signals (Divergence, Trough Agreement) trigger once and persist until the next cycle reset.

## RSI Reversal Formula

For each RSI period (21, 36, 56), the engine tracks the cycle low since the trough detection event. The cycle low is the minimum value that RSI has reached since the current scoring cycle began. The recovery ratio measures how far RSI has recovered from that low toward the 100 level, using a power-curve transformation that rewards early recovery more aggressively than late recovery. This means the first 30% of recovery from the cycle low earns a disproportionate share of the available points, which is exactly the behavior needed for early detection.

**cycle\_low** = min(RSIx) since trough

**recovery\_ratio** = (current\_RSI - cycle\_low) / (100 - cycle\_low)

**points** = max\_pts \* min(1.0, recovery\_ratio ^ 0.7)

If RSI drops below previous cycle low, points decrease proportionally. The power-curve exponent of 0.7 ensures that 50% recovery earns approximately 62% of max points, front-loading the reward for early movers.

## RSI Recovery Momentum Formula

This is a new v3 signal designed specifically for the Scout Engine. Instead of measuring absolute RSI level (which is what Reversal does), Momentum measures the **rate of change** in RSI. A stock whose RSI21 is rising at 2 points per bar from a cycle low of 35 is showing stronger recovery acceleration than one whose RSI21 is rising at 0.5 points per bar from a cycle low of 40. The momentum signal captures this distinction and rewards it with points, even when the absolute RSI level is still well below 50.

$\text{momentum} = (\text{RSI21\_current} - \text{RSI21\_3bars\_ago}) / 3$   
 $\text{positive\_threshold} = 0.5$  (points per bar)  
 $\text{points} = \text{max\_pts} * \min(1.0, \max(0, \text{momentum} - 0.5) / 2.0)$   
 Points scale linearly from 0 at threshold to max at momentum = 2.5. Negative momentum scores zero. This is evaluated per-timeframe independently.

## Multi-Speed RSI Divergence

An evolution of the standard RSI divergence signal. Instead of binary detection (divergence yes/no), this signal weights the divergence by how many of the three RSI periods (21, 36, 56) are simultaneously showing bullish divergence at the same local price minimum. If only RSI21 shows divergence, the signal earns a fraction of the maximum. If all three RSIs show divergence, it earns the full weight. This graduated approach reflects the higher conviction when multiple RSI speeds agree on the divergence pattern, while still rewarding single-speed divergence as an early-warning signal that precedes broader confirmation.

## Pool B - Price Structure | 25 pts

Pool B evaluates price structure for signs of basing and recovery. In the v2 matrix, Pool B was heavily weighted at 35 points, with the dominant signals being Price > TP28 (13 pts) and TP28 > WC50 (14 pts). Both of these are **late-firing** signals that only activate after price has already broken above key moving averages, which typically coincides with or follows the breakout price level. The Scout Engine redistributes Pool B weight to 25 points and replaces "Price > TP" with "Price Approaching TP," a proximity-based signal that begins scoring points while price is still below the TP28 line.

## Signal Table

| Signal                      | Condition                                                                                                                       | 4H  | D   | W   | Total | Lifecycle |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-------|-----------|
| <b>TP Flatten</b>           | TP28 range < 10% over 10 bars. Indicates basing/consolidation.                                                                  | 1.5 | 1.0 | 0   | 2.5   | Live      |
| <b>WC Flatten</b>           | WC50 range < 10% over 10 bars. Longer-term basing.                                                                              | 1.5 | 1.0 | 0   | 2.5   | Live      |
| <b>Price Approaching TP</b> | NEW. Proximity to TP28, scaled: within 10% = 40%, within 5% = 70%, within 2% = 100%. Above TP = 100%. Replaces v2 "Price > TP". | 2.5 | 4.0 | 5.0 | 11.5  | Live/die  |
| <b>TP &gt; WC</b>           | TP28 above WC50. Structural bullish alignment.                                                                                  | 2.0 | 3.0 | 3.5 | 8.5   | Live      |

*Raw maximum: 25.0 points. Weekly timeframe has no TP Flatten or WC Flatten signal by design, as weekly structural basing is captured indirectly through Pool C confirmation signals. The "Price Approaching TP"*

signal is the critical innovation here: by rewarding proximity rather than requiring price to be above the TP28 line, it allows the Scout Engine to accumulate up to 70% of the signal's points while price is still 5% below the TP28, creating meaningful scoring potential before the breakout.

## Price Approaching TP - Proximity Scaling

This signal replaces the v2 "Price > TP28" which required Close to be above the TP28 line before awarding any points. The new proximity-based approach recognizes that a stock trading at 95% of its TP28 is structurally very different from one trading at 70% of TP28, even though neither has crossed the line yet. The scaling formula uses a piecewise-linear approach with three thresholds, ensuring that the signal provides increasing reward as price converges on the TP28 level. Once price crosses above TP28, the signal awards full points.

```
distance_pct = (TP28 - Close) / TP28 * 100
If distance_pct <= 2%: scale = 1.0 (full points)
If distance_pct <= 5%: scale = 0.7
If distance_pct <= 10%: scale = 0.4
If distance_pct > 10%: scale = 0.0 (no points)
If Close > TP28: scale = 1.0 (full points)
points = max_pts_TF * scale
```

## Pool C - Confirmation | 15 pts

Pool C carries the lightest weight in the Scout Engine at 15 points, reflecting its role as a secondary confirmation layer rather than a primary scoring driver. In the v2 matrix, Pool C had 25 points including a 10-point Weekly Divergence signal and a 5-point D-RSI21 > 60 confirmation. The Scout Engine reduces these weights and lowers the D-RSI confirmation threshold from 60 to 50, allowing it to fire earlier in the recovery cycle. The 3D Divergence signal from v2 has been removed entirely since the 3-Day timeframe is no longer part of the data pipeline.

The philosophy here is deliberate: the Scout Engine prioritizes early detection over confirmation. A stock that scores 75+ on Pool A + Pool B alone represents a strong early-recovery candidate. Pool C provides a small additional boost when higher-timeframe confirmation arrives, but it is not required to reach the entry threshold. The Sentinel Engine, by contrast, will weight Pool C heavily to ensure that only confirmed trends receive full allocation.

| Signal                 | Condition                                                                                 | Pts | Lifecycle     | Notes                                                                       |
|------------------------|-------------------------------------------------------------------------------------------|-----|---------------|-----------------------------------------------------------------------------|
| <b>W Divergence</b>    | Weekly bullish divergence: price makes lower low, RSI21 makes higher low at local minima. | 5.0 | Fire-and-hold | Highest single-signal weight in Pool C. Persistent until next cycle reset.  |
| <b>D-RSI21 &gt; 50</b> | Daily RSI21 crosses above 50. Lowered from v2's >60 threshold for earlier activation.     | 3.0 | Live/die      | Fires when RSI enters neutral zone. Deactivates if RSI falls back below 50. |

| Signal                           | Condition                                                                                                                                                                              | Pts | Lifecycle     | Notes                                                                                     |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------|-------------------------------------------------------------------------------------------|
| <b>Cross-TF Boost</b>            | 2+ TFs in Pool A have active crossover (RSI21 > RSI36 or RSI21 > RSI56) AND Price Approaching TP scoring > 0 on any TF.                                                                | 4.0 | Live/die      | Multi-TF agreement bonus. Requires both RSI crossover alignment AND structural proximity. |
| <b>Multi-TF Trough Agreement</b> | Troughs detected on 2+ TFs within a 5-bar window. This signal is shared with Pool A and is listed here for classification clarity; it scores 2.5 pts allocated entirely within Pool A. | 2.5 | Fire-and-hold | Also counted in Pool A. Listed for reference. Actual points allocated in Pool A only.     |

*Pool C standalone maximum: 12.0 points (W Divergence 5 + D-RSI21 > 50 3 + Cross-TF Boost 4). The Multi-TF Trough Agreement is listed for classification completeness but its 2.5 points are allocated within Pool A. Combined Pool A + Pool B + Pool C maximum = 45 + 25 + 12 = 82 points from unique signals. With Pool A's full 45 pts including the 2.5 Trough Agreement, the core cap of 85 is approached but rarely reached in practice.*

## Bonus Pool | 25 pts

The Bonus Pool provides additive points on top of the core 85-point cap. These are structural bonuses that reward external factors beyond RSI recovery and price structure: volume patterns, proximity to support levels, secular trend quality, and the specific advantage of entering during an uptrend correction. Bonus points can push the total score above 85, with a maximum possible score of 110 (85 core + 25 bonus). Position sizing scales with the total score including bonus, meaning a stock with a 90 total (e.g., 70 core + 20 bonus) receives a larger allocation than one with a 75 total (e.g., 70 core + 5 bonus).

### Bonus A - Structural | 12 pts

| Signal                | Condition                                                                                                                                                       | Pts | Lifecycle | Notes                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|-------------------------------------------------------------------------------|
| <b>Volume Surge</b>   | Volume vs 20-bar average on Daily TF. Scaled: >1.5x = 2 pts, >2.5x = 4 pts, >3.5x = 5 pts.                                                                      | 5.0 | Live/die  | Confirms institutional buying interest during recovery.                       |
| <b>Support Bounce</b> | Proximity to 180-bar structural peak (prior swing high). Within 25% = 1 pt, within 15% = 2 pts, within 10% = 3 pts. Requires >= 15% decline from peak.          | 3.0 | Live/die  | Price holding near prior structural support increases recovery probability.   |
| <b>Trend Quality</b>  | Composite score (max 4 pts): Relative Strength vs sectoral index (1Y=1, 6M=0.5, 3M=0.5), MA Structure (1), Swing Lows Rising (1), Correction Compression (0.5). | 4.0 | Live      | Measures secular uptrend conviction. Uses best-matching Nifty sectoral index. |

### Bonus B - Uptrend Correction | 13 pts

Bonus B captures a specific high-probability scenario: stocks in a confirmed secular uptrend that correct by 25% or more represent superior entry opportunities because the long-term trend provides a tailwind for recovery. This sub-pool is gated by "Uptrend Confirmed" as a structural prerequisite. Other Bonus B signals cannot accumulate unless the stock is first classified as being in a secular uptrend (200-DMA rising and price above 200-DMA). This is not an inter-pool gate; it is an internal sub-pool prerequisite that only affects Bonus B points. Core pools A, B, C and Bonus A score independently regardless of uptrend status.

| Signal                      | Condition                                                                                                       | Pts | Lifecycle | Notes                                                                            |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------|-----|-----------|----------------------------------------------------------------------------------|
| <b>Uptrend Confirmed</b>    | Stock in secular uptrend: 200-DMA rising (slope > 0 over 50 bars) AND price above 200-DMA.                      | 2.0 | Live      | Prerequisite for this sub-pool. Required before other Bonus B signals activate.  |
| <b>Correction Depth</b>     | Price fallen >= 25% from recent swing high. Scaled: 25-35% = 3 pts, 35-45% = 4 pts, 45%+ = 5 pts.               | 5.0 | Live/die  | Deeper corrections in uptrends have higher recovery probability.                 |
| <b>RSI Turning Positive</b> | Any RSI (21/36/56) crossing above 50 from below. Per-TF: 4H = 0.5, D = 1.0, W = 1.0. Max 2.5 if all three fire. | 2.5 | Live/die  | Per-timeframe scoring. Rewards RSI crossing into neutral zone during correction. |
| <b>Price Above Support</b>  | Price holding above the 180-bar structural peak support level identified in Bonus A.                            | 2.0 | Live      | Confirms correction is holding at a structural level.                            |
| <b>Volume on Recovery</b>   | Above-average volume (>= 1.2x 20-bar avg) on up days during the recovery phase. Measured over last 5 up-days.   | 1.5 | Live/die  | Confirms institutional participation in the recovery.                            |

*Bonus B total:  $2 + 5 + 2.5 + 2 + 1.5 = 13.0$  points (when Uptrend Confirmed). If uptrend is not confirmed, Bonus B contributes 0 points. Bonus A scores independently in all cases. Total bonus pool maximum:  $12 + 13 = 25.0$  points.*

## Entry Decision Thresholds

The entry decision framework uses total score (core + bonus) to determine both the action and the position size. The thresholds are calibrated for the Scout Engine's early-detection profile, where a score of 75 represents strong multi-timeframe recovery evidence even if price has not yet broken above key moving averages. This is fundamentally different from the Sentinel Engine, where a 75 score would require structural confirmation that price is in a confirmed uptrend.

| Score Range    | Action       | Position Size | Rationale                                                                                                                    |
|----------------|--------------|---------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>0 - 49</b>  | DO NOT ENTER | 0%            | Insufficient recovery evidence. RSI may be basing but no momentum or structural confirmation.                                |
| <b>50 - 69</b> | PILOT BUY    | 25 - 30%      | Early recovery detected. Limited exposure while monitoring for confirmation. Watch for Pool A dynamic signals to strengthen. |

| Score Range | Action              | Position Size | Rationale                                                                                                                              |
|-------------|---------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 70 - 79     | STANDARD ENTRY      | 60 - 70%      | Solid multi-TF recovery with structural proximity. Scout Engine's primary entry zone. Most signals active but confirmation incomplete. |
| 80 - 89     | FULL ENTRY          | 100%          | Strong recovery across multiple domains. Core pools near maximum with bonus support. High-conviction early entry.                      |
| 90+         | FULL ENTRY (Strong) | 100%          | Maximum signal agreement with bonus. Rare for Scout Engine; typically requires both core near-cap and substantial bonus points.        |

The Pilot Buy zone (50-69) is particularly important for the Scout Engine. It represents the earliest point at which a position can be initiated, typically when RSI recovery is underway on 1-2 timeframes and price is approaching (but has not yet reached) the TP28 line. The position is intentionally small (25-30%) to manage risk while the remaining signals develop. If the score subsequently rises to 70+, the position is scaled up to Standard or Full Entry. If the score falls back below 50, the Pilot Buy is exited. This graduated approach ensures that the Scout Engine's early signals are actionable without requiring the full confirmation set that the Sentinel Engine demands.

*Maximum possible score: 110 (85 core + 25 bonus). In practice, scores above 90 are rare for the Scout Engine because it is designed to fire early, before all confirmation signals have activated. A score of 80-85 with bonus pushing it to 90+ is the typical ceiling scenario, requiring near-perfect alignment across all pools and timeframes plus substantial bonus support.*

## Trough-Gated Architecture

The entire entry scoring cycle is initiated by a trough detection event on the Daily timeframe. This defines when scoring begins and when all recovery baselines are established. Without a valid trough, the Scout Engine does not score the stock. This mechanism prevents the system from generating spurious signals during sustained uptrends or choppy, directionless markets where RSI oscillates without forming a meaningful cycle low. The trough acts as the anchor point from which all RSI recovery measurements are tracked.

| Parameter                 | Value                 | Description                                                                                                                             |
|---------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trough Detection</b>   | D-RSI21 <= 40         | 11-bar local minimum on Daily RSI21. The RSI21 value must be the lowest of the 5 bars on either side (11-bar window).                   |
| <b>Cycle Reset</b>        | On new trough         | When a new trough is detected, all recovery baselines (cycle_low) reset to the new RSI values. Previous recovery progress is discarded. |
| <b>Lookback (Support)</b> | 180 bars              | Window for identifying the prior structural peak used in Bonus A (Support Bounce) and Bonus B (Price Above Support).                    |
| <b>Support Reference</b>  | Prior structural peak | The swing high immediately preceding the highest peak within the 180-bar lookback. Not the last completed HH pattern top.               |

| Parameter             | Value            | Description                                                                                                              |
|-----------------------|------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Scoring Active</b> | Post-trough only | No scoring occurs before a valid trough is detected. Once a trough fires, scoring continues until the next trough reset. |

The 11-bar local minimum requirement ensures that the trough is not a transient spike but a genuine price-level where selling pressure has temporarily abated and RSI has found a floor. The  $\leq 40$  threshold on D-RSI21 ensures that the stock has experienced a meaningful correction rather than a minor pullback. For context, RSI21 values below 40 typically correspond to a 15-25%+ decline from recent highs in most Indian equities, which aligns well with the Bonus B uptrend correction requirement of 25%+ decline.

## Signal Lifecycle Definitions

Every signal in the Scout Engine has a defined lifecycle that determines how it activates, persists, and deactivates. Understanding these lifecycles is critical for interpreting the scoring matrix correctly, as two signals with identical point values can behave very differently depending on their lifecycle type. The lifecycle also affects how quickly a score can change: dynamic signals cause gradual score changes, while live/die signals can cause abrupt score shifts when conditions toggle.

| Lifecycle            | Behavior                                                                                                                                                            | Score Impact                                                                                                      | Example Signals                                          |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Dynamic</b>       | Points scale continuously with an underlying metric (RSI level, recovery ratio, momentum). No abrupt on/off. Score changes smoothly as the metric evolves.          | Gradual. Score moves up or down by small increments each bar as the metric changes. No sudden jumps.              | RSI21/36/56 Reversal, RSI Recovery Momentum              |
| <b>Live/die</b>      | Signal activates when condition is met, deactivates when condition fails. Acts as a binary switch with hysteresis to prevent flickering.                            | Abrupt. Full points awarded or removed when condition toggles. Can cause 5-10 point score swings in a single bar. | RSI21 > RSI36, Price Approaching TP, All RSI > 50        |
| <b>Fire-and-hold</b> | Signal triggers once when condition is first met and persists until the next cycle reset (new trough). Does not deactivate even if the original condition reverses. | One-time jump. Points are added once and remain for the entire scoring cycle. No subsequent reduction.            | RSI Divergence, Multi-TF Trough Agreement, W Divergence  |
| <b>Live</b>          | Similar to live/die but without deactivation. Once the condition is met, points persist as long as the condition remains true. No hysteresis.                       | Stable once activated. Points remain as long as the underlying condition holds. No flickering protection.         | TP Flatten, WC Flatten, Uptrend Confirmed, Trend Quality |

*Hysteresis for live/die signals uses a 2-bar sustained failure rule: the signal does not deactivate until the condition has failed for 2 consecutive bars. This prevents a single-bar RSI dip below a crossover from causing a premature deactivation. For example, if RSI21 dips below RSI36 for one bar and then recovers, the crossover*

*signal remains active. Only if RSI21 stays below RSI36 for 2 or more bars does the signal deactivate and its points are removed.*

# Design Principles

## No Inter-Pool Gating

Each pool scores independently based on its own conditions. Pool B does not require Pool A signals to activate, and Pool C does not require Pool A or Pool B scores to unlock. Grouping into A/B/C is purely organizational: Pool A captures momentum and RSI dynamics, Pool B captures price structure and basing patterns, and Pool C captures higher-timeframe confirmation. Any valid signal from any domain contributes its full weight without being suppressed or amplified by conditions in unrelated domains. The only exception is the internal Bonus B sub-pool, which requires "Uptrend Confirmed" as a prerequisite for its own signals; this does not affect any other pool or bonus.

## Per-Timeframe Independence

Each RSI signal is evaluated independently on each of the three timeframes. A bullish RSI21 > RSI36 crossover on the 4H chart scores its 1.5 points regardless of whether the Daily or Weekly crossover has occurred. Lower timeframes naturally lead higher timeframes in recovery: 4H signals fire first, followed by Daily, then Weekly. This cascading behavior is a core feature, not a bug, because it allows the Scout Engine to detect recovery in its earliest stages on the most sensitive timeframe. The per-TF independence also means that a single timeframe showing strong recovery can contribute meaningful points even if the other timeframes are lagging.

## Timeframe Weighting

Higher timeframes receive higher point values across all signals. A Weekly RSI divergence (2.0 pts) is worth 4x a 4H divergence (0.5 pts). This reflects the greater informational weight and rarity of higher-TF signals. A Weekly trough or crossover occurs far less frequently than a 4H equivalent, and when it does fire, it carries more conviction about the underlying trend change. The weighting ratios (4H : D : W) are approximately 1 : 1.7 : 2.3 across most Pool A signals, creating a natural progression that rewards multi-timeframe alignment without penalizing early 4H signals.

## Early-Firing Signal Philosophy

The defining characteristic of the Scout Engine is its preference for signals that fire early in the recovery cycle. This is achieved through three mechanisms. First, proximity-based signals (Price Approaching TP) reward approach rather than requiring crossing. Second, rate-based signals (RSI Recovery Momentum) reward the speed of recovery rather than the absolute level. Third, reduced confirmation thresholds (D-RSI21 > 50 instead of > 60) lower the bar for activation. The trade-off is that early signals are inherently noisier than late signals; a stock approaching its TP28 may fail to break through, and an

RSI showing positive momentum may reverse. The Pilot Buy zone (50-69) with its 25-30% position size is designed to manage this risk explicitly.

## Sectoral Relativity

Relative Strength in Bonus A (Trend Quality) is measured against the best-matching Nifty sectoral index, not a fixed benchmark or the broad Nifty 50. A defence stock is compared to Nifty India Defence, an IT stock to Nifty IT, a banking stock to Nifty Bank, and so on. This sector-relative comparison ensures that the Trend Quality score reflects genuine stock-specific strength rather than sector-wide momentum. A stock that outperforms its sectoral index by 10% over one year while the sector itself is flat is more impressive than a stock that rises 20% in a sector that rose 25%. The matching index is determined manually per stock based on its primary business segment and listed exchange classification.

## Consolidated Weight Summary

The table below provides a complete view of the Scout Engine scoring structure, showing every signal, its pool allocation, maximum possible points, lifecycle type, and whether it is a new v3 addition or carried over from v2. This serves as the definitive reference for implementation and backtesting.

| Pool              | Signal                      | Max Pts | Lifecycle     | v3 New?                   |
|-------------------|-----------------------------|---------|---------------|---------------------------|
| A                 | Multi-Speed RSI Divergence  | 3.5     | Fire-and-hold | Enhanced                  |
| A                 | RSI21 Reversal              | 7.5     | Dynamic       | No                        |
| A                 | RSI36 Reversal              | 3.0     | Dynamic       | No                        |
| A                 | RSI56 Reversal              | 2.0     | Dynamic       | No                        |
| A                 | RSI21 > RSI36               | 7.5     | Live/die      | No                        |
| A                 | RSI21 > RSI56               | 5.0     | Live/die      | No                        |
| A                 | RSI36 > RSI56               | 3.0     | Live/die      | No                        |
| A                 | RSI Recovery Momentum       | 7.0     | Dynamic       | Yes                       |
| A                 | Cross-TF Recovery Alignment | 4.0     | Live/die      | Yes                       |
| A                 | Multi-TF Trough Agreement   | 2.5     | Fire-and-hold | Yes                       |
| <b>A Subtotal</b> |                             | 45.0    |               |                           |
| B                 | TP Flatten                  | 2.5     | Live          | No                        |
| B                 | WC Flatten                  | 2.5     | Live          | No                        |
| B                 | Price Approaching TP        | 11.5    | Live/die      | Yes (replaces Price > TP) |
| B                 | TP > WC                     | 8.5     | Live          | No                        |

| Pool                    | Signal                              | Max Pts | Lifecycle     | v3 New?              |
|-------------------------|-------------------------------------|---------|---------------|----------------------|
| <b>B Subtotal</b>       |                                     | 25.0    |               |                      |
| <b>C</b>                | W Divergence                        | 5.0     | Fire-and-hold | No (reduced from 10) |
| <b>C</b>                | D-RSI21 > 50                        | 3.0     | Live/die      | Modified (was > 60)  |
| <b>C</b>                | Cross-TF Boost                      | 4.0     | Live/die      | No (reduced from 5)  |
| <b>C Subtotal</b>       |                                     | 12.0    |               |                      |
| <b>Bonus A</b>          | Volume Surge                        | 5.0     | Live/die      | No                   |
| <b>Bonus A</b>          | Support Bounce                      | 3.0     | Live/die      | No                   |
| <b>Bonus A</b>          | Trend Quality                       | 4.0     | Live          | Modified (reduced)   |
| <b>Bonus A Subtotal</b> |                                     | 12.0    |               |                      |
| <b>Bonus B</b>          | Uptrend Confirmed                   | 2.0     | Live          | No                   |
| <b>Bonus B</b>          | Correction Depth                    | 5.0     | Live/die      | No                   |
| <b>Bonus B</b>          | RSI Turning Positive                | 2.5     | Live/die      | Modified             |
| <b>Bonus B</b>          | Price Above Support                 | 2.0     | Live          | No                   |
| <b>Bonus B</b>          | Volume on Recovery                  | 1.5     | Live/die      | No                   |
| <b>Bonus B Subtotal</b> |                                     | 13.0    |               |                      |
| <b>TOTAL</b>            | Core Cap: 85   Bonus: 25   Max: 110 | 110     |               |                      |